Faheem Kamel

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Summary

Final-year Computer Science student at Singapore Management University, graduating May 2027. Sole engineer on a multi-tenant SaaS platform built from scratch, with a focus on tenant isolation, idempotent processing and tested releases.

Professional Experience

PCA Company Services – *Project Manager, Operis (ERP v2)* **Aug 2026 – Present**

- Run delivery of a roughly 40-module multi-tenant ERP built on a fork of the large open-source OpenMercato codebase, across four one-week sprints and two QA cycles, coordinating several teams and agreeing integration scope with stakeholders.
- Extending Model Context Protocol (MCP) and LLM integration across the platform, building on earlier AI tooling.

PCA Company Services – *Software Engineering & DevOps Intern* **May 2026 – Jul 2026**

- Sole engineer on a production-ready multi-tenant SaaS ERP built from scratch for launch in Vietnam: 8 business modules (incl. invoicing with government e-invoice ingestion), 352 REST API endpoints and 225,000+ lines of TypeScript.
- Engineered a directed-acyclic-graph workflow automation engine with distributed run leasing, a transactional outbox for domain events and four layers of idempotency, guaranteeing exactly-once side effects and automatic recovery of stalled runs.
- Designed multi-tenant PostgreSQL isolation across 12 schemas and 153 versioned migrations, enforcing tenant scoping on every read and write, and migrated authentication to Single Sign-On (ZITADEL OpenID Connect, OAuth 2.0 with PKCE), removing all credential storage from the application.
- Built asynchronous job infrastructure on Redis and BullMQ (7 queues with cron scheduling) and automated an 8-stage GitHub Actions CI/CD pipeline running 2,862 tests, migration replay and container vulnerability scanning before Dockerised deployment behind nginx.
- Designed the module system (per-module schemas, published cross-module interfaces, lint-enforced import boundaries) and reviewed other engineers' code; two engineers later shipped three more modules without touching the core.
- Built the React and TypeScript front end on a shared design system, with TanStack Query for server state and route-level code splitting so customers never download modules they cannot access.

INSPIRE Program (UVIC) – *Full Stack Developer Co-op — Action Hub* **Sep 2025 – Dec 2025**

- Developed a Model Context Protocol (MCP) server and workflow system that auto-injects context into LLM pipelines, supports multi-tenant usage and scales on AWS (Elastic Beanstalk, S3).
- Architected a context-aware AI email assistant grounded in meeting transcripts, CRM records and user uploads, powered by 8+ tools including semantic search, and hardened against prompt injection.
- Built an AI meeting intelligence platform on GPT-4 (notetakers, transcription, summaries, action items) with sub-100 ms SSE/WebSocket updates and 99.9% sync across Google Meet, Zoom and Teams.

Ensign InfoSecurity – *Full Stack Developer Intern* **Mar 2025 – Aug 2025**

- Architected and shipped an insider-threat investigation web application (Next.js, NestJS, PostgreSQL) integrating the MITRE ATT&CK framework, reducing investigation time by 40%.
- Automated ingestion and processing of 1,000+ CSV/Parquet records at 99%+ accuracy with duplicate detection and error handling, removing around 15 hours of manual work each week.
- Implemented Role-Based Access Control (RBAC) with case-level scoping, and containerised the application with Docker multi-stage builds for consistent releases across environments.

Education

Singapore Management University – *B.Sc. Computer Science* **Aug 2023 – May 2027**

- Concentrations: Software Development, Data Analysis, Machine Learning, Cybersecurity.

Technical Skills

Languages: TypeScript, JavaScript, Java, Python, SQL, C, HTML/CSS
Backend & Data: NestJS, Node.js, REST APIs, PostgreSQL, Redis, BullMQ, Prisma ORM, MongoDB, WebSockets
Infrastructure: Docker, GitHub Actions, CI/CD, nginx, AWS (S3, EC2, Elastic Beanstalk), Linux
Frontend: React, Next.js, React Native, TanStack Query, TailwindCSS
Security & Auth: OAuth 2.0, OpenID Connect (OIDC), Single Sign-On (SSO), RBAC, JWT, OWASP Top 10
AI & Practices: Model Context Protocol (MCP), GPT-4 integration, AI agents and tool use, Jest, Vitest, Git, Agile, JIRA

Awards & Leadership

Singapore Police Force – Valedictorian, 183rd intake; four Commander's Awards (Project Team), 2021 – 2023
Friends2Gather Community Service – Mentor, Jan 2024 – Dec 2024